Mathematical Comparison of the Two Solid-Velocity Interpolation Models

`laplaceAnchored` versus `laplaceSetValues`

Derived directly from `lambdaDotModel.C`

1 Purpose and notation

The code first converts discrete DEM particle velocities into a cell-centred raw velocity field $\boldsymbol{U}^{\mathrm{DEM}}$, and then constructs an interpolated Eulerian solid velocity field $\boldsymbol{U}$. Both vector equations are solved component by component for U_x, U_y, and U_z.

For cell i:

$$\mathcal{P}_i = \{\text{DEM particles whose centres lie in cell } i\}, \tag{1}$$

$$n_i = |\mathcal{P}_i|, \tag{2}$$

$$\boldsymbol{U}_i^{\mathrm{DEM}} = \begin{cases} \dfrac{1}{n_i} \displaystyle\sum_{p \in \mathcal{P}_i} \boldsymbol{v}_p, & n_i > 0.5, \\ \mathbf{0}, & n_i \le 0.5. \end{cases} \tag{3}$$

The symbols used below are:

Symbol	Meaning
ε_i	`porosityF` in cell i
ε_c	`solidPorosityCutoff`
V_i	volume of cell i
o_i	occupancy indicator, $o_i = \mathbb{I}(n_i > 0.5)$
$f = (i, j)$	face shared by neighbouring cells i and j
A_f	face area
d_{ij}	centre-to-centre distance normal to face f
D	Laplace diffusivity in `laplaceSetValues`; in the code $D = 10^{-4}\,\mathrm{m}^2$
c_{DEM}	strong anchor coefficient; default 10^6
c_{bg}	weak background anchor coefficient; default 10^{-12}

2 Step-by-step mathematical comparison

Step	`laplaceAnchored`	`laplaceSetValues`
1. Initial field	The solution field is initialized by $$\boldsymbol{U}_i^{(0)} = \boldsymbol{U}_i^{\mathrm{DEM}}.$$ Cells classified as non-solid are immediately reset to zero.	The solution field is initialized by $$\boldsymbol{U}_i^{(0)} = \boldsymbol{U}_i^{\mathrm{DEM}}.$$ No explicit gas-cell reset is applied before the Laplace solve.
2. Occupied-cell indicator	$$o_i = \begin{cases} 1, & n_i > 0.5, \\ 0, & n_i \leq 0.5. \end{cases}$$ Thus a stationary particle with $\boldsymbol{U}_i^{\mathrm{DEM}} = \boldsymbol{0}$ is still recognized as occupied.	The hard-reference set is determined from velocity magnitude: $$\mathcal{A} = \left\{ i : \left\lVert \boldsymbol{U}_i^{\mathrm{DEM}} \right\rVert > 0 \right\}.$$ Therefore an occupied cell whose mean DEM velocity is exactly zero is *not* included in the hard-reference set.
3. Solid-domain indicator	$$s_i = \mathbb{I}(\varepsilon_i < \varepsilon_c \ \vee \ o_i = 1).$$ An occupied cell is treated as solid even when its porosity is above the cutoff.	$$\chi_i = \mathrm{pos}(1 - \varepsilon_i).$$ Hence the Laplace region is inferred from whether ε_i is below unity, rather than directly from n_i.
4. Characteristic cell length	$$\ell_i^2 = \max\left(V_i^{2/3}, \varepsilon_{\mathrm{num}}\right).$$ This scales the anchor coefficient with mesh size.	No cell-length scaling is used in the main Laplace equation. The diffusivity D is spatially constant.
5. Anchor coefficient	First define $$c_i = \begin{cases} c_{\mathrm{DEM}}, & o_i = 1 \text{ or } s_i = 0, \\ c_{\mathrm{bg}}, & s_i = 1 \text{ and } o_i = 0. \end{cases}$$ Then $$a_i = \frac{c_i}{\ell_i^2}.$$ Thus occupied and gas cells receive a large reaction coefficient, while empty solid cells receive an almost-zero coefficient.	There is no reaction/penalty coefficient. Instead, cells in $\mathcal{A}$ are imposed through an algebraic hard constraint: $$\boldsymbol{U}_i = \boldsymbol{U}_i^{\mathrm{DEM}}, \qquad i \in \mathcal{A}.$$
6. Source field	Because empty DEM cells already have zero velocity, $$\boldsymbol{U}_i^{\mathrm{src}} = \begin{cases} \boldsymbol{U}_i^{\mathrm{DEM}}, & o_i = 1, \\ \boldsymbol{0}, & o_i = 0. \end{cases}$$ Gas cells are also assigned $\boldsymbol{U}_i^{\mathrm{src}} = \boldsymbol{0}$.	The main equation has no volumetric source: $$\boldsymbol{S}_i = \boldsymbol{0}.$$ The only prescribed nonzero information enters through `setValues`.

Step	`laplaceAnchored`	`laplaceSetValues`
7. Face diffusion mask	For an internal face $f=(i,j)$, the code first interpolates s to the face and then explicitly sets the coefficient to zero whenever either adjacent cell is non-solid. For the binary mask this is equivalent to $$\gamma_f = s_i s_j.$$ Therefore $$\gamma_f = \begin{cases} 1, & s_i = s_j = 1, \\ 0, & \text{otherwise.} \end{cases}$$	The cell mask is interpolated: $$\chi_f = \mathcal{I}_f(\chi).$$ An interface face is identified by $$0 < \chi_f < 1.$$ For those faces the matrix coupling is manually removed. This is equivalent to an effective face diffusivity $$D_f^{\text{eff}} = \begin{cases} 0, & 0 < \chi_f < 1, \\ D, & \text{otherwise.} \end{cases}$$
8. Continuous governing equation	The code solves a screened-Laplace or reaction-diffusion equation: $$-\nabla \cdot (\gamma \nabla \boldsymbol{U}) + a\boldsymbol{U} = a\boldsymbol{U}^{\text{src}}. \tag{A}$$	Away from hard-reference cells, the code solves the homogeneous Laplace equation: $$\nabla \cdot (D \nabla \boldsymbol{U}) = \mathbf{0}. \tag{B}$$ Since the right-hand side is zero, reversing the overall matrix sign does not change the mathematical solution.
9. Finite-volume face weight	For a coupled solid-solid face $f=(i,j)$: $$w_{ij}^A = \gamma_f \frac{A_f}{d_{ij}}.$$ If either cell is non-solid, $w_{ij}^A = 0$.	For a face that is not cut: $$w_{ij}^B = D \frac{A_f}{d_{ij}}.$$ For a solid-gas interface face, $w_{ij}^B = 0$.
10. Cell equation before constraints	Ignoring explicit non-orthogonal corrections, cell i satisfies $$\left(a_i V_i + \sum_{j \in N(i)} w_{ij}^A \right) \boldsymbol{U}_i - \sum_{j \in N(i)} w_{ij}^A \boldsymbol{U}_j = a_i V_i \boldsymbol{U}_i^{\text{src}}. \tag{A-FV}$$	For an unconstrained cell: $$\left(\sum_{j \in N(i)} w_{ij}^B \right) \boldsymbol{U}_i - \sum_{j \in N(i)} w_{ij}^B \boldsymbol{U}_j = \mathbf{0}. \tag{B-FV}$$
11. Treatment of DEM cells	A DEM cell is a *soft* anchor during the linear solve: $$a_i \gg 1 \quad \Longrightarrow \quad \boldsymbol{U}_i \approx \boldsymbol{U}_i^{\text{DEM}}.$$ The equality is not algebraically exact inside the matrix solve.	A DEM cell in $\mathcal{A}$ is a *hard* anchor. Its matrix row is replaced by $$\boldsymbol{U}_i = \boldsymbol{U}_i^{\text{DEM}}.$$ Thus the prescribed value is exact during the main Laplace solve.

Step	`laplaceAnchored`	`laplaceSetValues`
12. Treatment of empty solid cells	For $s_i = 1$ and $o_i = 0$: $$a_i = \frac{c_{\mathrm{bg}}}{\ell_i^2} \approx 0.$$ Therefore $$-\nabla \cdot (\gamma \nabla \boldsymbol{U}) \approx \mathbf{0},$$ so the field is nearly harmonic, with a very weak tendency toward zero.	There is no background reaction term: $$\nabla \cdot (D \nabla \boldsymbol{U}) = \mathbf{0}.$$ The value is a purely harmonic extension of the hard-reference values and boundary information.
13. Treatment of gas cells	For $s_i = 0$: $$\gamma_f = 0 \quad \text{on adjacent internal faces,} \qquad a_i = \frac{c_{\mathrm{DEM}}}{\ell_i^2}, \qquad \boldsymbol{U}_i^{\mathrm{src}} = \mathbf{0}.$$ Thus $$a_i \boldsymbol{U}_i = \mathbf{0} \quad \Longrightarrow \quad \boldsymbol{U}_i \approx \mathbf{0}.$$	The solid-gas matrix coupling is removed, corresponding to $$\boldsymbol{n} \cdot D \nabla \boldsymbol{U} = 0$$ at the internal solid-gas interface. Gas cells are not hard-set to zero during the main solve, but they are zeroed in the final post-processing step when $\varepsilon_i \geq \varepsilon_c$.
14. Boundary/interface condition	The zero face coefficient imposes an internal no-diffusion condition: $$\boldsymbol{n} \cdot \gamma \nabla \boldsymbol{U} = 0$$ between solid and non-solid cells.	The removed face coefficients impose $$\boldsymbol{n} \cdot D \nabla \boldsymbol{U} = 0$$ between the $\chi = 1$ and $\chi = 0$ regions. The code also removes the corresponding coupling on processor boundaries.
15. Linear-solver repetitions	The equation is solved $$N_A = \texttt{nUsInterpolationCorrectors}$$ times. Before each solve, equation relaxation is applied.	The main hard-constrained Laplace system is solved up to five times. Let $\boldsymbol{r}_0^{(k)}$ be the initial residual vector at repetition k. The stopping test is $$\left\| \boldsymbol{r}_0^{(k)} \right\| < 10^{-3}$$ and $$\left| \frac{\left\| \boldsymbol{r}_0^{(k-1)} \right\|}{\max(\left\| \boldsymbol{r}_0^{(k)} \right\|, \varepsilon_{\mathrm{num}})} - 1 \right| < 10^{-1}.$$

Step	`laplaceAnchored`	`laplaceSetValues`
16. Optional corrector	No separate pseudo-transient equation is used. Repeated solves apply the same anchored steady equation (A).	When $N_B = \texttt{nLaplaceSetValuesCorrectors} > 0,$ the code solves $c_t \dfrac{\partial \boldsymbol{U}}{\partial t} - \nabla \cdot (D \nabla \boldsymbol{U}) = \boldsymbol{0}, \qquad c_t = 10^{-4}\,\mathrm{s}, \qquad \text{(B-corr)}$ for N_B repetitions. In a first-order finite-volume form: $\dfrac{c_t V_i}{\Delta t}\left(\boldsymbol{U}_i^{m+1} - \boldsymbol{U}_i^{m}\right) + \sum_j w_{ij}^{B}\left(\boldsymbol{U}_i^{m+1} - \boldsymbol{U}_j^{m+1}\right) = \boldsymbol{0}.$ The `setValues` constraints are not re-applied inside this corrector system.
17. Final overwrite	After the solve: $\boldsymbol{U}_i \leftarrow \begin{cases} \boldsymbol{U}_i^{\mathrm{DEM}}, & n_i > 0.5, \\ \boldsymbol{0}, & n_i \le 0.5 \text{ and } \varepsilon_i \ge \varepsilon_c, \\ \boldsymbol{U}_i, & \text{otherwise.} \end{cases} \qquad \text{(A-post)}$	The same final overwrite is applied: $\boldsymbol{U}_i \leftarrow \begin{cases} \boldsymbol{U}_i^{\mathrm{DEM}}, & n_i > 0.5, \\ \boldsymbol{0}, & n_i \le 0.5 \text{ and } \varepsilon_i \ge \varepsilon_c, \\ \boldsymbol{U}_i, & \text{otherwise.} \end{cases} \qquad \text{(B-post)}$ Thus occupied and gas-cell outputs are made exact after either method.
18. Mathematical character	**Soft constrained screened Laplace:** diffusion + reaction penalty. The anchor strength is finite and mesh-scaled.	**Hard constrained Laplace:** pure harmonic extension with exact internal Dirichlet values imposed by replacing matrix rows.

3 The essential mathematical difference

The two equations can be written compactly as

$$\texttt{laplaceAnchored:} \qquad \mathcal{L}_A(\boldsymbol{U}) = -\nabla\cdot(\gamma\nabla\boldsymbol{U}) + a(\boldsymbol{U}-\boldsymbol{U}^{\mathrm{src}}) = \boldsymbol{0}, \tag{4}$$

$$\texttt{laplaceSetValues:} \qquad \mathcal{L}_B(\boldsymbol{U}) = \nabla\cdot(D\nabla\boldsymbol{U}) = \boldsymbol{0}, \qquad \boldsymbol{U}_i = \boldsymbol{U}_i^{\mathrm{DEM}} \quad (i\in\mathcal{A}). \tag{5}$$

Therefore:

- In `laplaceAnchored`, DEM information enters through a finite penalty $a_i(\boldsymbol{U}_i - \boldsymbol{U}_i^{\mathrm{DEM}})$.
- In `laplaceSetValues`, DEM information enters as an exact algebraic constraint $\boldsymbol{U}_i = \boldsymbol{U}_i^{\mathrm{DEM}}$.
- In an empty solid region, `laplaceAnchored` contains a tiny zero-directed reaction term, while `laplaceSetValues` is a pure Laplace equation.
- For $c_{\mathrm{DEM}} \to \infty$ and $c_{\mathrm{bg}} \to 0$, the anchored model approaches the hard-constrained Laplace model, provided that both methods use the same solid domain and the same reference cells.

4 One-dimensional interpretation

Consider a one-dimensional empty solid interval $0 < x < L$ between two known DEM velocities $\boldsymbol{U}_L$ and $\boldsymbol{U}_R$.

`laplaceSetValues`

With constant D,

$$D\frac{\mathrm{d}^2\boldsymbol{U}}{\mathrm{d}x^2} = \boldsymbol{0}, \qquad \boldsymbol{U}(0) = \boldsymbol{U}_L, \qquad \boldsymbol{U}(L) = \boldsymbol{U}_R.$$

Hence

$$\boxed{\boldsymbol{U}(x) = \boldsymbol{U}_L + \frac{x}{L}\left(\boldsymbol{U}_R - \boldsymbol{U}_L\right)}$$

which is exactly linear.

`laplaceAnchored`

Inside an empty solid region with constant weak anchor a_{bg} and $\boldsymbol{U}^{\mathrm{src}} = \boldsymbol{0}$,

$$-\gamma\frac{\mathrm{d}^2\boldsymbol{U}}{\mathrm{d}x^2} + a_{\mathrm{bg}}\boldsymbol{U} = \boldsymbol{0}.$$

Define

$$\lambda^2 = \frac{a_{\mathrm{bg}}}{\gamma}.$$

Then

$$\boldsymbol{U}(x) = \boldsymbol{C}_1\cosh(\lambda x) + \boldsymbol{C}_2\sinh(\lambda x).$$

When a_{bg} is extremely small, $\lambda L \ll 1$, so

$$\cosh(\lambda x) \approx 1, \qquad \sinh(\lambda x) \approx \lambda x,$$

and the result becomes almost linear. It is not mathematically identical to the pure Laplace result unless $a_{\mathrm{bg}} = 0$ and the DEM anchors become exact.

5 Conditions under which both methods become nearly equivalent

The outputs approach each other when all of the following hold:

$$c_{\mathrm{DEM}} \to \infty, \tag{6}$$
$$c_{\mathrm{bg}} \to 0, \tag{7}$$
$$s_i = \chi_i \quad \text{for every cell}, \tag{8}$$
$$\{i : o_i = 1\} = \{i : \left\|\boldsymbol{U}_i^{\mathrm{DEM}}\right\| > 0\}, \tag{9}$$
$$N_B = 0, \tag{10}$$

and both methods use identical boundary conditions, mesh, numerical schemes, linear solvers, tolerances, relaxation, and parallel decomposition.

The fourth condition fails for a stationary particle:

$$o_i = 1 \quad \text{but} \quad \left\|\boldsymbol{U}_i^{\mathrm{DEM}}\right\| = 0.$$

In that case `laplaceAnchored` recognizes the cell as a DEM anchor, whereas `laplaceSetValues` does not anchor it during the main solve.

6 Compact comparison

Property	`laplaceAnchored`	`laplaceSetValues`
PDE type	Screened Laplace / reaction-diffusion	Pure Laplace
DEM constraint during main solve	Soft, finite penalty	Hard, exact row replacement
Empty-solid background bias	Tiny bias toward zero	No volumetric bias
Gas handling during solve	Strong zero anchor and disconnected diffusion	Interface disconnection; final gas reset
Stationary DEM particle	Recognized using n_i	Missed by $\left\|\boldsymbol{U}_i^{\mathrm{DEM}}\right\| > 0$ in main solve
Cell-size dependence	Anchor scales as $V_i^{-2/3}$	Constant D; mesh enters through FV geometry
Optional pseudo-time step	No	Yes
Final occupied-cell value	Exactly $\boldsymbol{U}_i^{\mathrm{DEM}}$	Exactly $\boldsymbol{U}_i^{\mathrm{DEM}}$

7 Implementation note

The model-selection line in the supplied file is written as

```
lookupOrDefault<word>("laplaceAnchored", "interpolationMode").
```

For the usual OpenFOAM argument order `(keyword, defaultValue)`, the intended form appears to be

```
lookupOrDefault<word>("interpolationMode", "laplaceAnchored")
```

Otherwise the code may look for a dictionary keyword named `laplaceAnchored` and use `interpolationMode` as the default value.